

Implementation of Molten Salt Thermophysical Properties in MARS-KS for Molten Salt Reactor Thermal-Hydraulic Analysis

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ABSTRACT

Molten salt reactors (MSRs) have attracted considerable attention as Generation IV reactor concepts due to the safety characteristics, high operating temperatures, and potential for efficient fuel utilization. However, additional thermophysical property models and heat transfer correlations are required to accurately represent the thermal-hydraulic behavior of molten salt systems within conventional system analysis codes. In this study, a molten salt thermal-hydraulic analysis framework was developed using the MARS-KS code to enable the analysis of both fluoride based and chloride based molten salt systems. Thermophysical property correlations for FLiBe, FLiNaK, and NaCl-KCl-MgCl₂ were implemented, including density, specific heat capacity, thermal conductivity, viscosity, and surface tension. In addition, specific internal energy and entropy correlations were generated using Maxwell relations and incorporated into the existing MARS-KS thermodynamic framework. To improve the prediction of convective heat transfer behavior, the Gnielinski correlation was adopted for the calculation of heat transfer coefficients. The developed methodology was assessed through steady-state simulations of the Molten Salt Reactor Experiment (MSRE). Comparison with reference MSRE data demonstrated reasonable agreement for thermal-hydraulic parameters, including core temperatures, heat exchanger temperatures, and system pressures, with relative errors below 1%. Furthermore, the axial temperature distributions of the fuel salt and graphite moderator were consistently reproduced. The results demonstrate that the modified MARS-KS code can successfully represent the dominant thermal-hydraulic behavior of molten salt reactor systems and provide a unified analysis capability for both fluoride based and chloride based molten salts. The developed framework can serve as a foundation for thermal-hydraulic analysis of advanced molten salt reactor concepts.

KEYWORDS

Molten Salt Reactor, MARS-KS, Thermal Hydraulic Analysis, Thermophysical Properties

1. INTRODUCTION

Interest in Generation IV nuclear reactor systems has increased significantly due to the demand for enhanced safety, improved fuel utilization, and sustainable energy production [1]. Among various advanced reactor concepts, molten salt reactors (MSRs) have attracted considerable attention because of low-pressure operation, high operating temperature, favorable heat transfer characteristics, and the capability to achieve high thermal efficiency. The high boiling temperature and thermophysical properties of molten salts enable reactor operation at elevated temperatures without requiring high system pressure. Furthermore, the large

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volumetric heat capacity of molten salts provides efficient heat transport and decay heat removal capability, which can contribute to enhanced operational flexibility and passive safety characteristics. Despite the advantages, several technological challenges remain before the widespread deployment of MSRs can be achieved. Long-term exposure of structural materials to high temperature molten salt environments could result in corrosion, material degradation, and changes in system performance [2-3]. Furthermore, recent MSR development efforts have expanded beyond conventional fluoride-based concepts to include chloride-based reactor systems, leading to a wider range of coolant properties and thermal-hydraulic behaviors that must be accurately evaluated during design and safety analyses. Consequently, reliable thermal hydraulic analysis methodologies capable of representing the characteristics of various molten salt systems are essential for supporting reactor design, operational assessment, and safety evaluation. The establishment of analysis frameworks would be required for the development and licensing of future molten salt reactor technologies.

Therefore, several studies have been conducted to investigate the applicability of system analysis codes to molten salt reactor technologies and high-temperature molten salt systems. The evaluation of thermophysical and thermochemical properties of molten salts and the assessment of heat transfer correlations under molten salt operating conditions, composed of 60 wt% NaNO_3 and 40 wt% KNO_3 . The heat transfer predictions were validated utilizing the RELAP5-3D code [4]. In addition, the analysis and validation of the Molten Salt Reactor Experiment (MSRE), which remains one of the most important benchmark facilities for molten salt reactor research, were carried out. RELAP5-based analyses have been performed using point kinetics models that account for delayed neutron transport within both the reactor core and external circulation loop, enabling the simulation of thermal-hydraulic behavior under representative MSRE operating conditions [5]. Similarly, the system analysis module (SAM), developed under the Nuclear Energy Advanced Modeling and Simulation (NEAMS) program, has been applied to the MSRE primary loop and compared with RELAP5 predictions to evaluate its thermal-hydraulic modeling capability [6]. In addition to thermal-hydraulic analysis based on individual system codes, coupled multi-physics frameworks for molten salt reactor applications were investigated. Coupled TRACE–Serpent frameworks have been developed to analyze MSRE transients and evaluate reactor behavior under various operating scenarios [7]. Dynamic performance analysis was also conducted using RELAP5/MOD4.0, where system models and power control strategies were implemented to investigate load-following characteristics of molten salt reactors [8]. Furthermore, MSRE results involving inserted reactivity were analyzed using RELAP5-3D [9], while natural circulation heat removal performance under MSRE conditions has been evaluated using the GAMMA+ code [10]. The studies demonstrated that existing system analysis codes can reproduce important thermal-hydraulic and neutronic phenomena associated with molten salt reactor systems. Nevertheless, most previous investigations have focused on specific reactor concepts and individual molten salt compositions, particularly fluoride based salts. As advanced reactor development has expanded toward both fluoride and chloride based systems, significant variations in thermophysical properties and heat transfer characteristics. Therefore, the development of a generalized thermal-hydraulic analysis framework capable of accommodating multiple molten salt compositions within a single system analysis platform remains an important research objective.

In this study, the applicability of the MARS-KS system analysis code to molten salt reactor thermal-hydraulic analyses was investigated through the implementation of representative fluoride and chloride based molten salt properties. Thermophysical property models for FLiBe (LiF-BeF_2), FLiNaK (LiF-NaF-KF), and NaCl-KCl-MgCl₂ were incorporated into the code to extend its capability beyond conventional water-cooled reactor applications. These salts were selected because they represent major candidate coolants and fuel carrier salts currently being considered for thermal-spectrum and fast-spectrum molten salt reactor concepts. In addition to the implementation of molten salt thermophysical properties, heat transfer correlations suitable for molten salt flow conditions were incorporated to improve the prediction of convective heat transfer. Since the thermophysical characteristics of molten salts differ substantially from those of water, including thermal conductivity, viscosity, specific heat capacity, and Prandtl number

behavior, the applicability of conventional heat transfer models requires assessment. Therefore, modifications were conducted to reflect the heat transfer performance in molten salt systems. The developed analysis framework was applied to simulations of the MSRE. Steady-state thermal-hydraulic characteristics predicted by the modified MARS-KS code were compared with available MSRE data to assess the capability of the implemented models. Thermal-hydraulic analysis framework capable of accommodating both fluoride and chloride based molten salts within a unified system analysis code environment could provide a basis for future analysis of advanced molten salt reactor concepts.

2. THERMAL PROPERTIES OF MOLTEN SALTS

Thermophysical property models for FLiBe, FLiNaK, and NaCl-KCl-MgCl₂ were implemented in the MARS-KS code to enable thermal-hydraulic analyses of representative fluoride- and chloride-based molten salt systems. Since MARS-KS was originally developed for conventional water-cooled reactor applications, additional property correlations were required to represent the thermodynamic behavior of molten salts over the operating temperature range of interest. Correlations for density, specific heat capacity, thermal conductivity, and dynamic viscosity were collected from the literature and implemented as functions of temperature. The thermodynamic properties required by the code, including internal energy and entropy, were derived from the implemented property correlations using Maxwell relations and fundamental thermodynamic identities. The formulations enabled the direct utilization of molten salt properties within the existing MARS-KS. The resulting property database provides a unified thermodynamic representation for FLiBe, FLiNaK, and NaCl-KCl-MgCl₂, allowing comparative thermal-hydraulic analysis to be performed within a system analysis environment. Furthermore, accurate estimation of convective heat transfer is essential for molten salt reactor analysis, considering the thermophysical characteristics of molten salts. Numerous previous studies have evaluated the applicability of existing heat transfer correlations to molten salt systems and reported that the Gnielinski correlation provides satisfactory agreement with experimental data over a wide range of molten salt compositions and operating conditions [4, 11]. Therefore, the Gnielinski correlation was adopted in this study for the calculation of convective heat transfer coefficients. The implemented correlation was subsequently applied to FLiBe, FLiNaK, and NaCl-KCl-MgCl₂ to improve the estimation capability of the thermal-hydraulic behavior under representative molten salt operating conditions. Equations (1) and (2) represent the Darcy friction factor required for the correlation and the Nusselt number: where f is the Darcy friction factor, Re is the Reynolds number, Pr is the Prandtl number, h is the convective heat transfer coefficient, k is the thermal conductivity of the molten salt, and D_h is the hydraulic diameter of the flow channel.

$$f = [0.79 * \ln(Re) - 1.64]^{-2} \quad (1)$$

$$Nu = \frac{\left(\frac{f}{8}\right)(Re-1000)Pr}{1+12.7\left(\frac{f}{8}\right)^{1/2}(Pr^{2/3}-1)} \quad (2)$$

2.1. FLiBe

FLiBe has been widely considered as a reference molten salt for thermal spectrum molten salt reactor concepts because of neutronic characteristics, chemical stability, and extensive experimental database. Consequently, FLiBe was selected as the first candidate salt for implementation. To accurately represent the thermal-hydraulic behavior of FLiBe within the MARS-KS, thermophysical property correlations were obtained from the literature and incorporated into the code [12]. The implemented properties include surface tension, thermal conductivity, viscosity, specific volume, and specific heat capacity. In addition, thermodynamic properties, including internal energy and entropy, were derived from the implemented property correlations using thermodynamic relationships. The correlations adopted for each property are summarized in Equations (3)–(9): where σ is the surface tension, k is the thermal conductivity, μ is the dynamic viscosity, v is the specific volume, C_p is the specific heat capacity, s is the specific entropy, U is

the specific internal energy, T is the temperature, and P is the pressure. T_{ref} and P_{ref} denote the reference temperature and pressure, respectively.

$$\sigma = 0.29578 - 1.2 * 10^{-4} * T \quad (3)$$

$$k = 0.6297 - 0.5 * 10^{-3} * T \quad (4)$$

$$\mu = 1.16 * 10^{-4} e^{\frac{3755}{T}} \quad (5)$$

$$v = \frac{1}{2413 - 0.488 * T} \quad (6)$$

$$C_p = 2415.78 \quad (7)$$

$$s = 2415.78 * \ln\left(\frac{T}{T_{ref}}\right) - \frac{0.488}{(2413 - 0.488 * T)^2} (P - P_{ref}) \quad (8)$$

$$U = 2415.78 * (T - T_{ref}) - \frac{0.488 * T}{(2413 - 0.488 * T)^2} (P - P_{ref}) - P \left(\frac{1}{2413 - 0.488 * T} - \frac{1}{2413 - 0.488 * T_{ref}} \right) \quad (9)$$

2.2. FLiNaK

FLiNaK is another representative fluoride based molten salt that has been widely considered for advanced reactor and high-temperature heat transport applications due to relatively low melting temperature compared with FLiBe. In addition, experimental measurements of thermophysical properties have been reported in the literature, leading to FLiNaK a suitable candidate for thermal-hydraulic model development and verification. Therefore, FLiNaK was selected as the second fluoride based molten salt for implementation in this study. Thermophysical property correlations for surface tension, thermal conductivity, dynamic viscosity, specific volume, and specific heat capacity were applied from the literature and incorporated into the MARS-KS code [12]. Consistent with the FLiBe implementation, specific entropy and internal energy were derived from the implemented property correlations using thermodynamic relationships to ensure compatibility with the MARS-KS thermodynamic framework. The correlations adopted for FLiNaK are summarized in Equations (10)–(16).

$$\sigma = 0.2726 - 1.014 * 10^{-4} * T \quad (10)$$

$$k = 0.36 - 5.6 * 10^{-4} * T \quad (11)$$

$$\mu = 4.0 * 10^{-5} e^{\frac{4170}{T}} \quad (12)$$

$$v = \frac{1}{2579.3 - 0.624 * T} \quad (13)$$

$$C_p = 976.78 + 1.0634 * T \quad (14)$$

$$s = 976.78 * \ln\left(\frac{T}{T_{ref}}\right) + 1.0634(T - T_{ref}) - \frac{0.624}{(2579.3 - 0.624 * T)^2} (P - P_{ref}) \quad (15)$$

$$U = 976.78 * (T - T_{ref}) + 0.5317(T^2 - T_{ref}^2) - \frac{0.624 * T}{(2579.3 - 0.624 * T)^2} (P - P_{ref}) - P \left(\frac{1}{2579.3 - 0.624 * T} - \frac{1}{2579.3 - 0.624 * T_{ref}} \right) \quad (16)$$

2.3. NaCl-KCl-MgCl₂

NaCl-KCl-MgCl₂ was considered as a representative chloride based molten salt to extend the applicability of the developed framework beyond conventional fluoride systems. Recently, chloride salts have attracted considerable interest for fast-spectrum molten salt reactor concepts. Therefore, temperature dependent correlations for surface tension, thermal conductivity, dynamic viscosity, specific volume, and specific heat capacity were implemented in MARS-KS [13]. The corresponding thermodynamic properties, including specific entropy and internal energy, were subsequently generated using the same methodology applied to the fluoride salts. The adopted property correlations are presented in Equations. (17)–(23).

$$\sigma = 0.1330 - 0.48 * 10^{-4} * T \quad (17)$$

$$k = 0.5822 - 2.6 * 10^{-4} * T \quad (18)$$

$$\mu = 0.706 * 10^{-5} e^{\frac{1204}{T}} \quad (19)$$

$$v = \frac{1}{1958.84 - 0.56355 * T} \quad (20)$$

$$C_p = 1301.38 + 0.5 * T \quad (21)$$

$$s = 1301.38 * \ln\left(\frac{T}{T_{ref}}\right) + 0.5(T - T_{ref}) - \frac{0.56355}{(1958.84 - 0.56355 * T)^2} (P - P_{ref}) \quad (22)$$

$$U = 1301.38 * (T - T_{ref}) + 0.25(T^2 - T_{ref}^2) - \frac{0.56355 * T}{(1958.84 - 0.56355 * T)^2} (P - P_{ref}) - P \left(\frac{1}{1958.84 - 0.56355 * T} - \frac{1}{1958.84 - 0.56355 * T_{ref}} \right) \quad (23)$$

3. COMPARISON WITH MSRE

Verification and assessment of the developed analysis framework are essential before its application to advanced molten salt reactor concepts. Although thermophysical property models and heat transfer correlations can be implemented within a system analysis code, the ability to reproduce the thermal-hydraulic behavior of molten salt systems should be demonstrated through comparison. The MSRE, operated at Oak Ridge National Laboratory, represents one of the most extensively documented molten salt reactor facilities and provides valuable operational data for code validation. Consequently, MSRE has been widely adopted as a benchmark problem for the evaluation of thermal-hydraulic and neutronic analysis tools. In the present study, simulations of the MSRE primary system were performed using the modified MARS-KS code to assess the applicability of the implemented molten salt property models and heat transfer correlations. Predicted thermal-hydraulic parameters were compared with available reference data to evaluate the performance of the developed methodology. Since the objective of this work is to establish a thermal-hydraulic analysis framework applicable to both fluoride and chloride based molten salts, confirmation of the implemented models is required. The capability of the implemented thermophysical property models and heat transfer can be assessed prior to application to advanced molten salt reactor concepts.

3.1. Configuration of MARS-KS analysis

To assess the applicability of the developed molten salt thermal-hydraulic analysis framework, steady-state simulations of the MSRE were performed using the modified MARS-KS code. Because detailed operating conditions and geometric information are available in the literature, the MSRE provides a suitable basis for

evaluating the implemented molten salt property models and heat transfer correlations. Figure 1 presents the nodalization scheme developed for the MSRE in MARS-KS. The one-dimensional hydraulic components are considered while preserving the principal geometric characteristics of the experimental facility. The nodalization was established to reproduce the overall flow path of the system and to reflect the heat transfer rate of primary, heat exchanger, and radiator sides occurring during reactor operation.

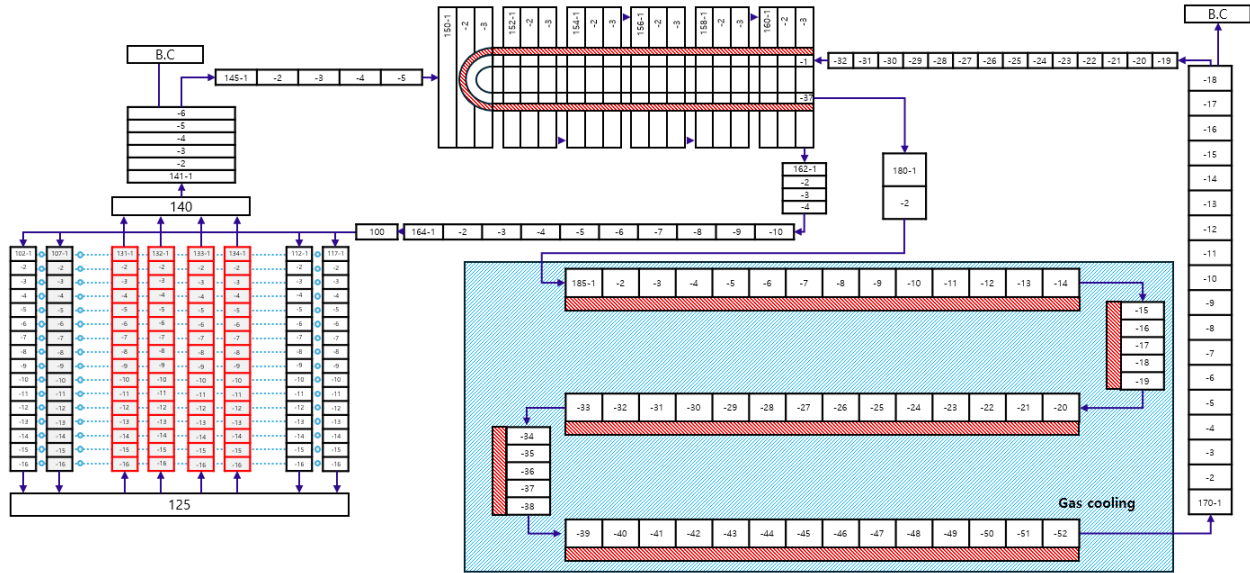


Figure 1. Schematic of MARS-KS nodalization for MSRE system

Table I. Operating conditions and geometric parameters used in the MSRE simulation

Operating parameter	Value	Unit
Core power	10	MW
Primary side pressure	2.413×10^5	Pa
Graphite channel hydraulic diameter	15.85	mm
Graphite channel height	1.73	m
Primary side flowrate	1200	gpm
Heat generation rate [Graphite]	6	%
Heat generation rate [Fuel]	94	%
Heat exchanger baffle distance	0.3048	m
Heat exchanger tube diameter	12.7	mm
Heat exchanger pressure	3.792×10^5	Pa
Heat exchanger flowrate	850	gpm
Effective heat transfer area of heat exchanger	24.06	m ²
Radiator tube diameter	19.05	mm
Radiator tube flowrate	200,000	cfm

Table 1 summarizes the principal operating and geometric parameters adopted in this analysis. The reactor thermal power was specified as 10 MW, corresponding to the nominal operating condition of the MSRE [14]. Additional parameters, including hydraulic diameter, coolant mass flow rate, and inlet temperature, were obtained from available design documentation and incorporated into the MARS-KS model.

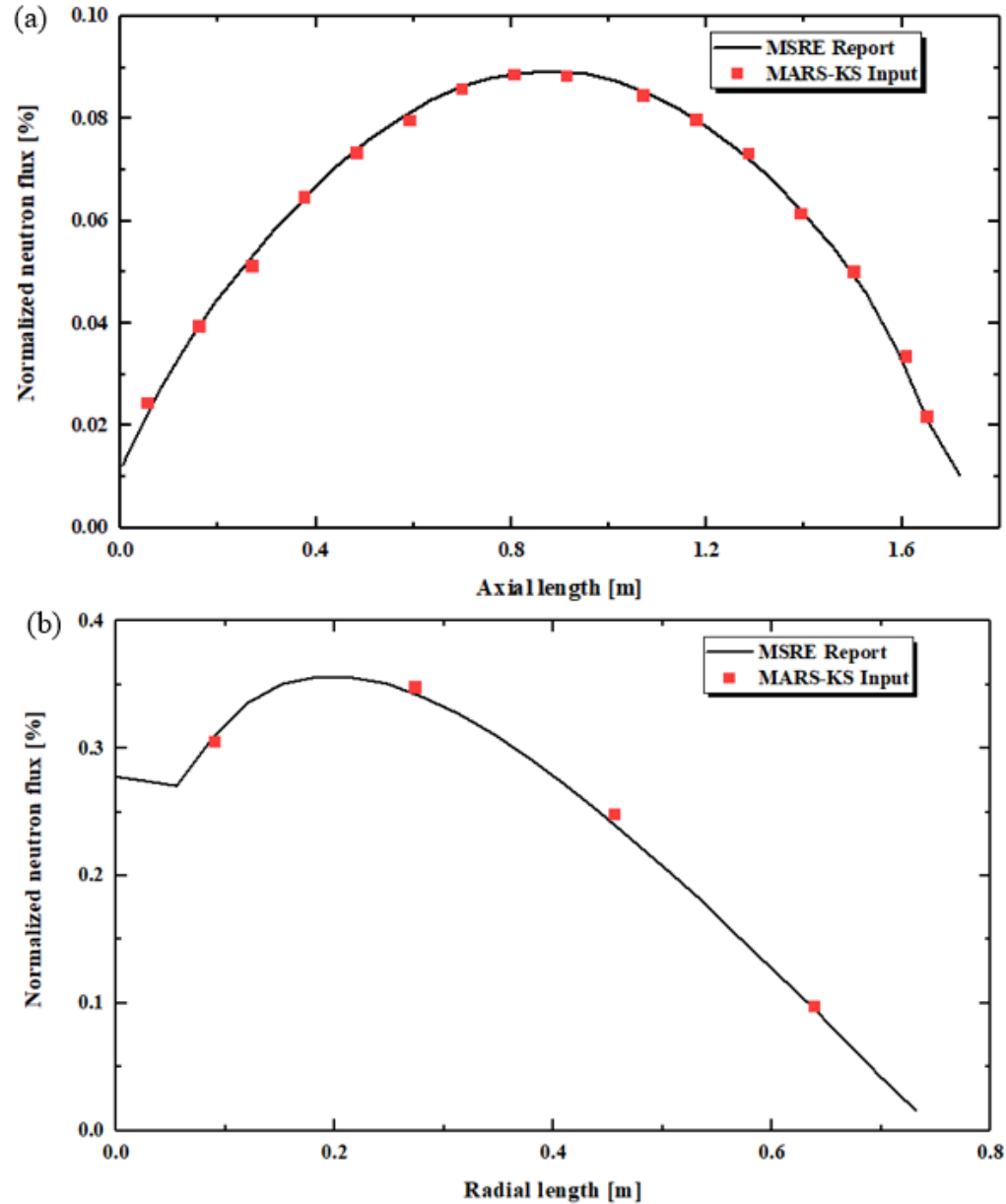


Figure 2. Normalized neutron flux distribution of MSRE and MARS-KS simulation: (a) axial and (b) radial

Since heat generation within the reactor core affects coolant temperature distribution and thermal-hydraulic behavior, a power distribution was incorporated into the simulation. Figure 2 presents the normalized axial and radial power distributions adopted in the present study [15]. The power distribution was applied in both the axial and radial directions to represent the non-uniform neutron flux and corresponding heat generation

characteristics within the reactor core. The heat source distribution was implemented in the core region of the MARS-KS and utilized throughout the steady-state simulations. The use of a normalized power distribution allows the spatial variation of heat generation to be represented more realistically than a uniform power assumption. Therefore, local temperature gradients and coolant heating characteristics within the core can be predicted with improved fidelity, leading to a more representative assessment of the thermal-hydraulic behavior under nominal MSRE operating conditions.

3.2. Results of MARS-KS analysis

Table 2 summarizes the comparison between the reference MSRE operating conditions and the steady-state estimation obtained using the modified MARS-KS code. Overall, the developed analysis framework reproduced the thermal-hydraulic characteristics of the MSRE with relative errors below 1% for all evaluated parameters. The predicted core inlet and outlet temperatures were 911.77 K and 937.55 K, respectively, corresponding to relative errors of 0.40% and 0.17%. In addition, the heat exchanger inlet and outlet temperatures were predicted within 0.19% and 0.83% of the reference values. The agreement in temperature predictions indicates that the implemented thermophysical property correlations and heat transfer model accurately reflect the heat transfer phenomena. Since temperature distributions in molten salt systems are dependent on density, specific heat capacity, thermal conductivity, and viscosity, the results demonstrate the consistency of the developed molten salt property database within the MARS-KS framework. Similarly, the pressure showed reasonable agreement, with a deviation of only 0.6%. Overall, the comparison demonstrates that the modified MARS-KS code is capable of reproducing the thermal-hydraulic behavior of the MSRE. The agreement observed for both temperature and pressure parameters provides confidence in the applicability of the developed framework for thermal-hydraulic analyses of fluoride and chloride based molten salt reactor systems.

Table II. Comparison of MSRE and MARS-KS estimation under steady-state conditions

Operating parameter	MSRE	MARS-KS	Error [%]
Core inlet temperature	908.15 K	911.77 K	0.40
Core outlet temperature	935.93 K	937.55 K	0.17
Primary side pressure	2.413x10 ⁵ Pa	2.426x10 ⁵ Pa	0.54
Heat exchanger inlet temperature	824.82 K	823.29 K	0.19
Heat exchanger outlet temperature	866.48 K	859.33 K	0.83
Heat exchanger pressure	3.792x10 ⁵ Pa	3.795x10 ⁵ Pa	0.08

Figure 3 compares the axial temperature distributions of graphite moderator and fuel salt predicted by MARS-KS with the reference MSRE results. Both distributions exhibit the temperature increase according to the core height. The behavior predicted by MARS-KS is consistent with the reference solution, demonstrating that the imposed power distribution and implemented thermophysical properties adequately represent the thermal behavior of the reactor core. For the fuel salt, the temperature profile predicted by MARS-KS follows the reference distribution throughout the majority of the core region. The temperature difference remains relatively small, particularly in the central portion of the core, where the heat generation rate is highest. The agreement indicates that the implemented FLiBe thermophysical properties and convective heat transfer model reproduced the primary heat transfer. In contrast, deviations are observed for the graphite moderator temperature distribution. Although the overall behavior is reproduced, the peak graphite temperature predicted by MARS-KS is 2.5% lower than the reference value. Despite the differences, the fuel salt temperature distribution was reproduced, indicating that the dominant thermal-

hydraulic behavior of the reactor system was appropriately reflected. The maximum fuel temperature predicted by MARS-KS differed from the reference value by 7 K, corresponding to an error of 0.74%. The agreement demonstrates that the developed framework is capable of providing reliable estimates of coolant temperature under molten salt reactor operating conditions. Overall, the comparison against MSRE reference data confirms that the implemented thermophysical property database and Gnielinski-based heat transfer treatment can successfully reproduce the dominant thermal-hydraulic characteristics of molten salt reactor systems. The results suggest the applicability of the developed MARS-KS framework to both fluoride based and chloride based molten salt reactor concepts, by establishing a foundation for analysis involving alternative salt compositions and reactor configurations.

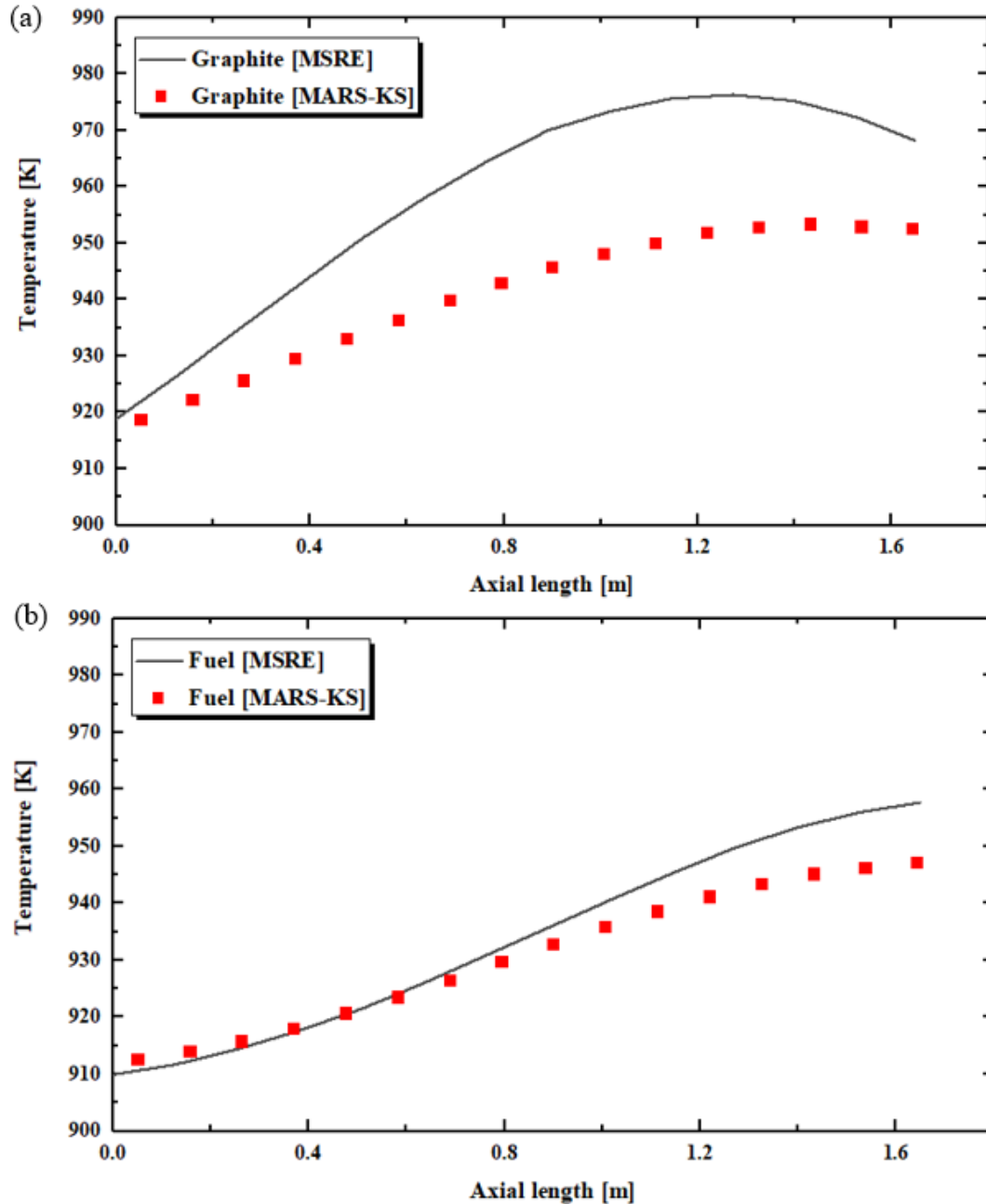


Figure 3. Comparison of axial temperature distributions between MSRE and MARS-KS predictions: (a) graphite moderator and (b) fuel salt

4. CONCLUSIONS

Interest in molten salt reactors has continued to increase due to enhanced safety, high-temperature operation, and improved fuel utilization. Therefore, reliable thermal-hydraulic analysis methods are required to support the design and safety evaluation of future molten salt reactor systems. However, the application of conventional system analysis codes to molten salt reactors remains limited by the availability of appropriate thermophysical property models and heat transfer correlations. In this study, a molten salt thermal-hydraulic analysis framework was developed using the MARS-KS code. Temperature-dependent thermophysical property correlations for FLiBe, FLiNaK, and NaCl-KCl-MgCl₂ were implemented to enable the analysis of representative fluoride-based and chloride-based molten salt systems. In addition, specific internal energy and entropy correlations were generated from the implemented property models using Maxwell relations and incorporated into the existing MARS-KS thermodynamic framework. To improve the prediction of convective heat transfer behavior in molten salt systems, the Gnielinski correlation was adopted and implemented for the calculation of heat transfer coefficients. The developed methodology was assessed through steady-state simulations of the MSRE. Comparison with reference MSRE data demonstrated agreement for major thermal-hydraulic parameters, including core temperatures, heat exchanger temperatures, and system pressures, with relative errors below 1%. Furthermore, the axial temperature distributions of both the fuel salt and graphite moderator were reasonably reproduced, confirming the capability of the modified MARS-KS code to represent the dominant thermal-hydraulic characteristics of molten salt reactor systems. Although the developed framework reproduced the thermal-hydraulic behavior of the MSRE, the present validation was restricted to FLiBe-based operating conditions. Since significant differences exist in thermophysical properties between fluoride and chloride based molten salts, additional validation against experimental data for FLiNaK and NaCl-KCl-MgCl₂ systems should be performed. Despite the limitation, the results demonstrate that the proposed framework can provide a unified thermal-hydraulic analysis capability for molten salts within MARS-KS. The developed methodology could serve as a foundation for analysis of advanced molten salt reactor concepts and can be extended to transient, accident, and system design studies involving a wide range of molten salt compositions and reactor configurations.

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REFERENCES

1. J. Serp, M. Allibert, O. Beneš, S. Delpech, O. Feynberg, V. Ghetta, D. Zhimin, et al., “The Molten Salt Reactor (MSR) in Generation IV: Overview and Perspectives,” *Progress in Nuclear Energy*, 77, pp. 308–319 (2014).
2. V. Ignatiev and A. Surenkov, “Corrosion Phenomena Induced by Molten Salts in Generation IV Nuclear Reactors,” *Structural Materials for Generation IV Nuclear Reactors*, pp. 153–189, Woodhead Publishing, Cambridge, United Kingdom (2017).
3. G.L. Fredrickson, *Molten Salt Reactors: Materials Challenges*, INL/CON-22-69122-Rev000, Idaho National Laboratory, Idaho Falls, Idaho, USA (2022).
4. S. Yoon, K. Frick, C. Davis, C. Parisi, and G. Mesina, “Assessment of RELAP5-3D Code with Molten Salt Heat Transfer Experiments,” *Nuclear Technology*, pp. 1–22 (2025).

5. C. Shi, M. Cheng, and G. Liu, "Development and Application of a System Analysis Code for Liquid Fueled Molten Salt Reactors Based on RELAP5 Code," *Nuclear Engineering and Design*, 305, pp. 378–388 (2016).
6. A.M. Leandro, F. Heidet, R. Hu, and N.R. Brown, "Thermal Hydraulic Model of the Molten Salt Reactor Experiment with the NEAMS System Analysis Module," *Annals of Nuclear Energy*, 126, pp. 59–67 (2019).
7. T. Hanusek and R.M. Juan, "Analysis of the Power and Temperature Distribution in Molten Salt Reactors with TRACE: Application to the MSRE," *Annals of Nuclear Energy*, 157, 108208 (2021).
8. G. Chen, M. Li, Y. Zou, and H. Xu, "Analysis of Load-Following Operation Characteristics of Liquid Fuel Molten Salt Reactor," *Progress in Nuclear Energy*, 150, 104308 (2022).
9. H. Mochizuki, "Validation of Neutronics and Thermal-Hydraulics Coupling Model of the RELAP5-3D Code Using the MSRE Reactivity Insertion Tests," *Nuclear Engineering and Design*, 389, 111669 (2022).
10. N.I. Tak, H.S. Lim, S.N. Lee, and S.J. Kim, "Simulation of Natural Convection in the Molten Salt Reactor Experiment Using GAMMA+," *Proceedings of the Korean Nuclear Society Autumn Meeting, Changwon, Republic of Korea, October 19–21, 2022*.
11. F. Haddad and P. Li, "Entropy Generation Minimization to Optimize Heat Transfer in CSP Technologies Using Molten Salt System NaCl/KCl/MgCl₂ as Heat Transfer Fluids," *Proceedings of ASME International Mechanical Engineering Congress and Exposition (IMECE 2021), Virtual Conference, November 1–5, 2021, V011T11A062* (2021).
12. M.S. Sohal, M.A. Ebner, P. Sabharwall, and P. Sharpe, *Engineering Database of Liquid Salt Thermophysical and Thermochemical Properties*, INL/EXT-10-18297, Idaho National Laboratory, Idaho Falls, Idaho, USA (2010).
13. C. Villada, W. Ding, A. Bonk, and T. Bauer, "Engineering Molten MgCl₂–KCl–NaCl Salt for High-Temperature Thermal Energy Storage: Review on Salt Properties and Corrosion Control Strategies," *Solar Energy Materials and Solar Cells*, 232, 111344 (2021).
14. R.C. Robertson, *MSRE Design and Operations Report, Part I: Description of Reactor Design*, Oak Ridge National Laboratory, Oak Ridge, Tennessee, USA (1965).
15. P.N. Haubenreich, J.R. Engel, B.E. Prince, and H.C. Claiborne, *MSRE Design and Operations Report, Part III: Nuclear Analysis*, ORNL-TM-730, Oak Ridge National Laboratory, Oak Ridge, Tennessee, USA (1964).